

Exercises

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Surname, First name
Statistics (BCS1520)
 BCS1520 Mock Exam

1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9
0	0	0	0	0	0	0

☐ a ☒ b ☐ c ☐ d ☐ e ☐ f → b
☐ a ☐ b ☒ c ☐ d ☐ e ☐ f → c
☒ a ☐ b ☐ c ☒ d ☐ e ☐ f → a

Answer multiple-choice questions as shown in the example.

Program: Bachelor Computer Science**Course code:** BCS1520**Examiners:** Marijn ten Thij and Tim Dick**Date/time:** N/A**Format:** Closed book**Allowed aids:** Pen, and simple calculator from DACS approved calculator list, formula sheet (provided)**Instructions to students:**

- The exam consists of 7 questions (plus question 8 for extra space for answers) on 14 pages.
- Fill in your name and student ID number on the cover page and tick the corresponding numerals of your student number in the table (top right cover page).
- Answer every question in the reserved space below the question. Do not write outside the reserved space or on the back of pages, this will not be scanned and will NOT be graded! As a last resort if you run out of space, use the extra answer space at the end of the exam.
- In no circumstance write on or near the QR code at the bottom of the page!
- Ensure that you properly motivate your answers.
- Only use black or dark blue pens, and write in a readable way. Do not use pencils.
- Answers that cannot be read easily cannot be graded and may therefore lower your grade.
- If you think a question is ambiguous, or even erroneous, and you cannot ask during the exam to clarify this, explain this in detail in the space reserved for the answer to the question.
- If you have not registered for the exam, your answers will not be graded, and thus handled as invalid.
- You are not allowed to have a communication device within your reach, nor to wear or use a watch.
- You have to return all pages of the exam. You are not allowed to take any sheets, even blank, home.
- The points in this exam sum up to 80.
- Good luck!

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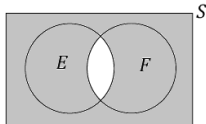
Multiple Choice Questions

- 2p **1a** Suppose that the province of Limburg contains 85 school corporations and each corporation reports its expenditures per pupil. If we want to construct a 95%-confidence interval for the average expenditure per pupil, which type of confidence interval (CI) is most appropriate in this situation?
- (a) The p -CI as we are dealing with a proportion.
 - (b) The z -CI as we know the population variance.
 - (c) The t -CI as we do not know the population variance.
 - (d) None, this sample is not a random sample and therefore constructing a CI is not appropriate.

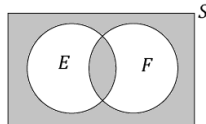
- 2p **1b** Which of the following visualizations highlights (indicated by the gray area) the following subset of the sample space S :

$$(E' \cap F') \cup (E \cap F)$$

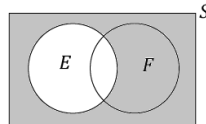
I



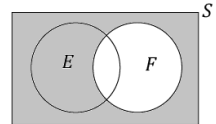
II



III



IV



- (a) IV
 - (b) III
 - (c) II
 - (d) I
- 2p **1c** The sampling distribution of the sample variance follows a
- (a) χ^2 distribution with $n - 1$ degrees of freedom
 - (b) t -distribution with $n - 1$ degrees of freedom
 - (c) χ^2 distribution with 1 degrees of freedom
 - (d) χ^2 distribution with n degrees of freedom
 - (e) t -distribution with n degrees of freedom

- 2p **1d** Fill in the blank: In Geary's test for normality, the ____ of the test statistic is a good estimator for σ if X follows a normal distribution with mean μ and variance σ^2 , but it will over- or underestimate for any other distribution.
- (a) denominator
 - (b) numerator
 - (c) first factor
 - (d) first term

Statement: The binomial distribution can **not** be approximated by a normal distribution.

1p

- a

5p

Please be as complete as possible in your answer. Remember that the only thing that can be graded is what you write down on your exam.

Statement: When performing a t -test, and obtaining a p -value of 0.07, then there is a 7% probability that the alternative hypothesis is false.

1p **2c** Is the statement True or False?

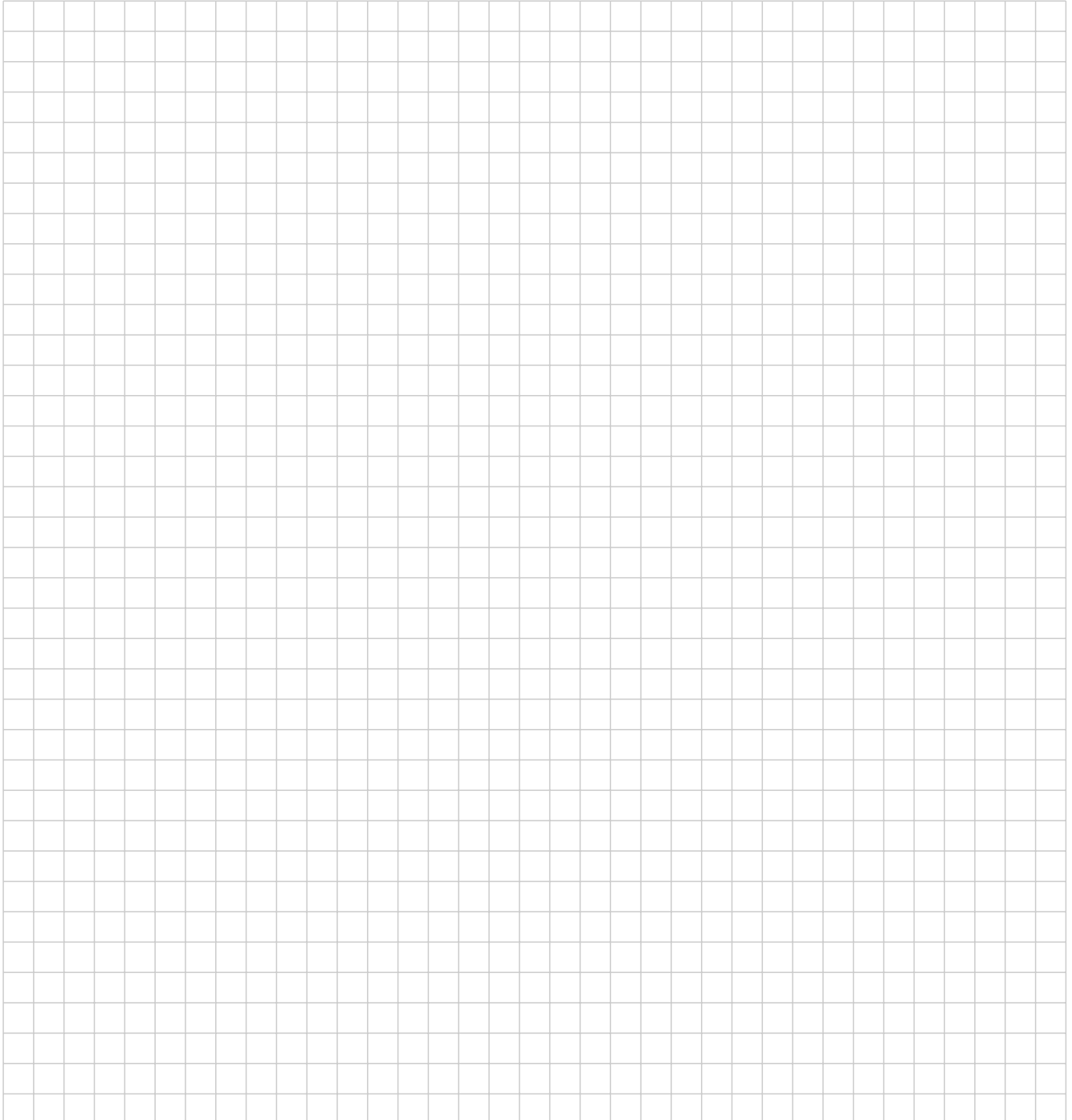
(a) True (b) False

5p **2d** Motivate your response to the previous question on whether the statement is True or False.

Please be as complete as possible in your answer. Remember that the only thing that can be graded is what you write down on your exam.

Random Drawing

- 10p **3** An urn contains b black balls and r red balls. One of the balls is drawn at random, but when it is put back in the urn c additional balls of the same color are put in with it. Now suppose that we draw another ball. Derive that the probability of the first ball drawn was black given that the second ball drawn was red is equal to $\frac{b}{b+r+c}$.

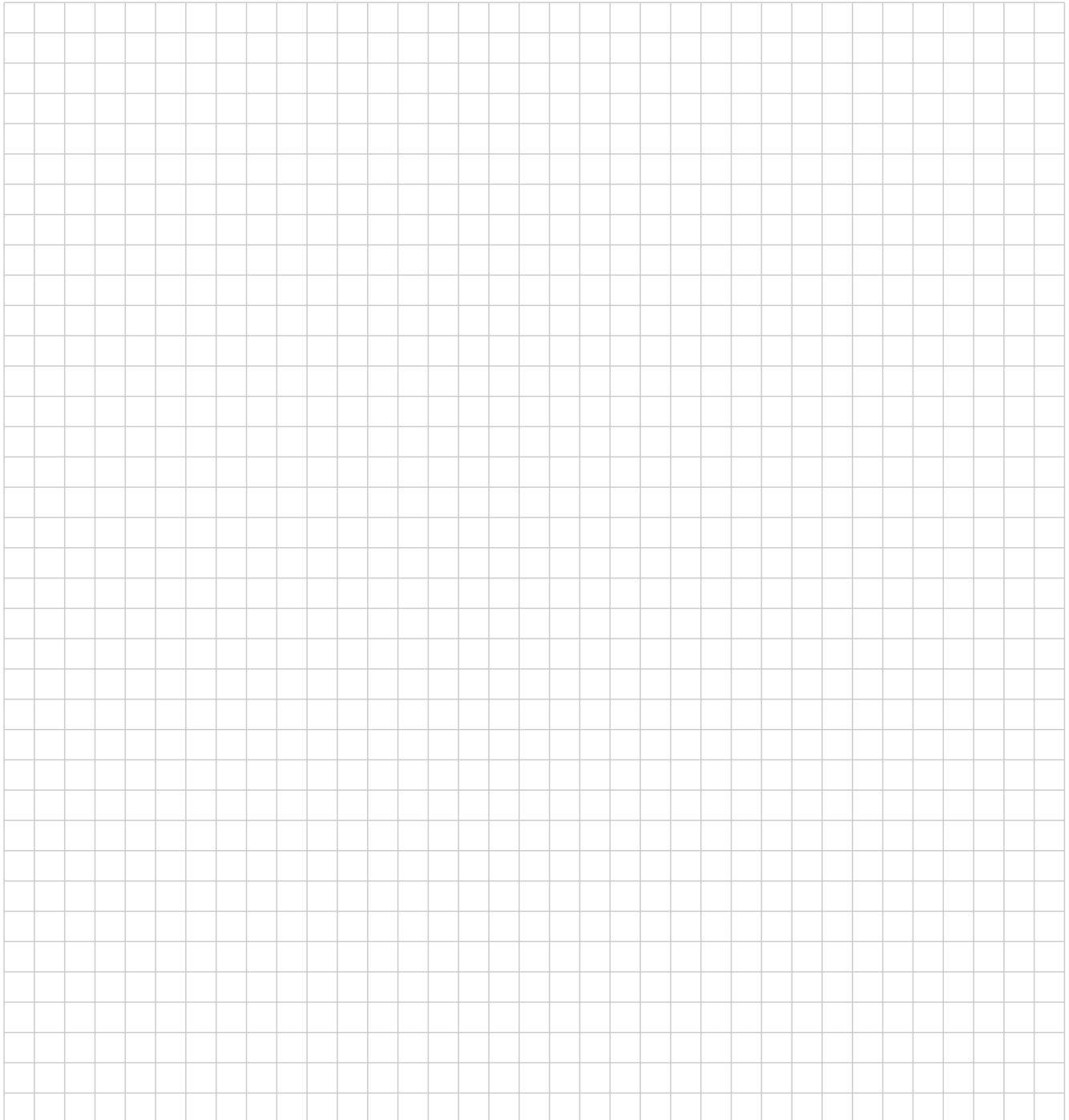


Joint Distributions

6p **4a** Consider the joint density function

$$f(x, y) = \begin{cases} \frac{16y}{x^3} & x > 2, 0 < y < 1, \\ 0 & \text{elsewhere.} \end{cases}$$

Compute the correlation coefficient ρ_{XY} .



A convenience store has two separate locations where customers can be checked out as they leave. These locations each have two cash registers and two employees who check out customers. Let X be the number of cash registers being used at a particular time for location 1 and Y the number being used at the same time for location 2. The joint probability function is given by

	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.12	0.04	0.04
$x = 1$	0.08	0.19	0.05
$x = 2$	0.06	0.12	0.30

4p **4b** Calculate $Var(X)$

4p **4c** Calculate $E[Y | X = 2]$

A retailer entered into an exclusive agreement with a supplier who guaranteed to provide all products at competitive prices. The retailer eventually began to purchase supplies from other vendors who offered better prices. The original supplier filed a legal action claiming violation of the agreement. In defense, the retailer had an audit performed on a random sample of invoices. For each audited invoice, all purchases made from other suppliers were examined and the prices were compared with those offered by the original supplier. For each invoice, the percent of purchases for which the alternate supplier offered a lower price than the original supplier was recorded. Here are the data:

6p **5a** Based on the information above, construct a 95% CI for the average percentage of the supplier's prices that are not competitive.

Researchers asked 180 women to read a hypothetical shopping experience where they entered a luxury store (e.g., Louis Vuitton, Gucci, Burberry) and ask a salesperson for directions to the items they seek. For half the women, the salesperson was condescending while doing this. The other half were directed in a neutral manner. After reading the experience, participants were asked what price they were willing to pay (in USD) for a particular product from the brand. The results are summarized in the table below.

Luxury Store	n	$\bar{x}$	s
Treatment: Condescending	90	4.44	3.98
Treatment: Neutral	90	3.95	2.88

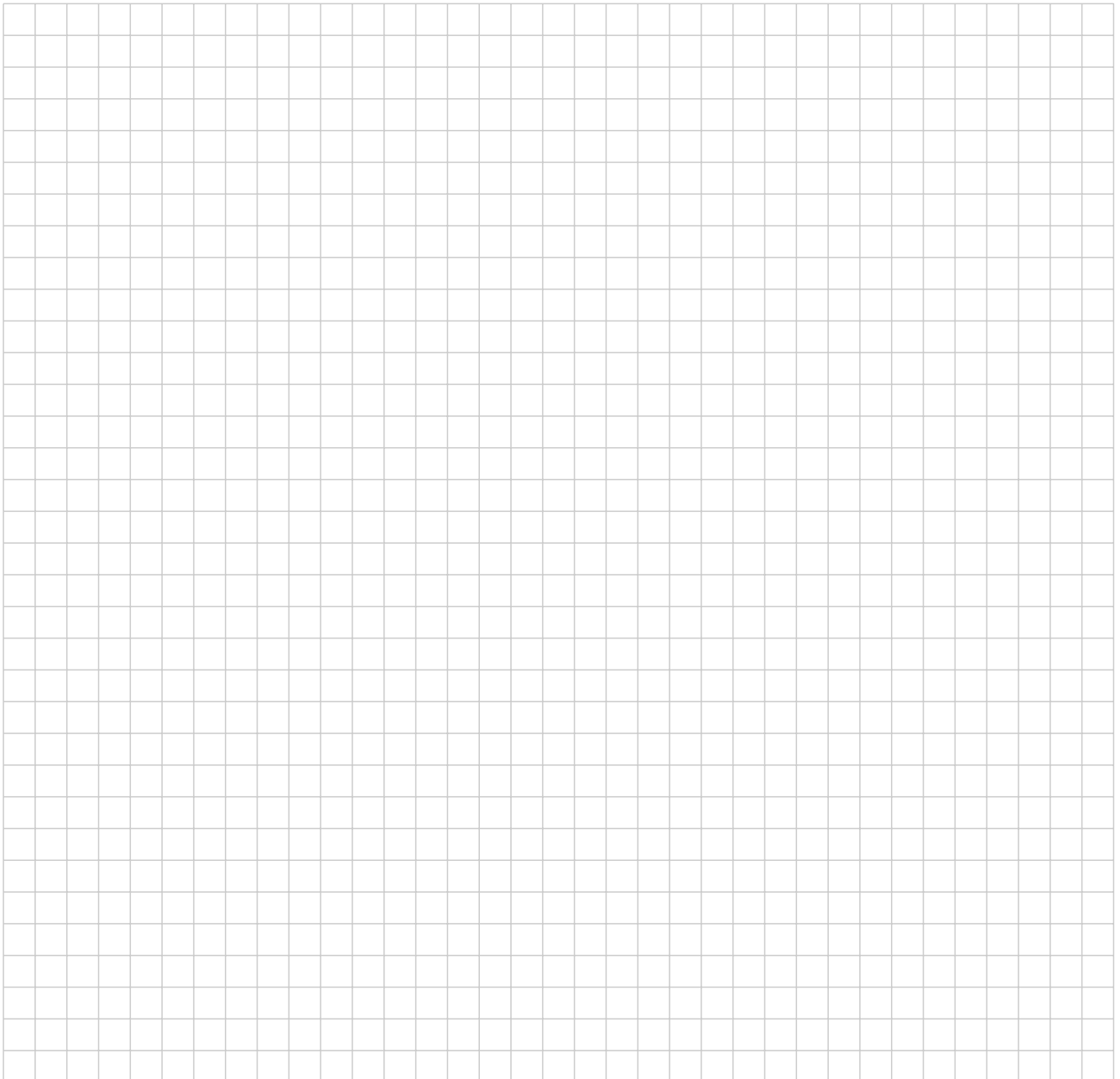
- 6p **6a** Were the participants who were treated rudely in the luxury store willing to pay more for the product? Use significance testing to determine the answer.



In another study the same researchers asked 180 different women to read the same hypothetical shopping experience, but now they entered a mass market (e.g., Gap, H&M). Again, after reading the experience, participants were asked what price they were willing to pay (in USD) for a particular product from the brand. The results are summarized in the table below.

Mass Market	n	$\bar{x}$	s
Treatment: Condensing	90	2.90	3.28
Treatment: Neutral	90	2.98	3.24

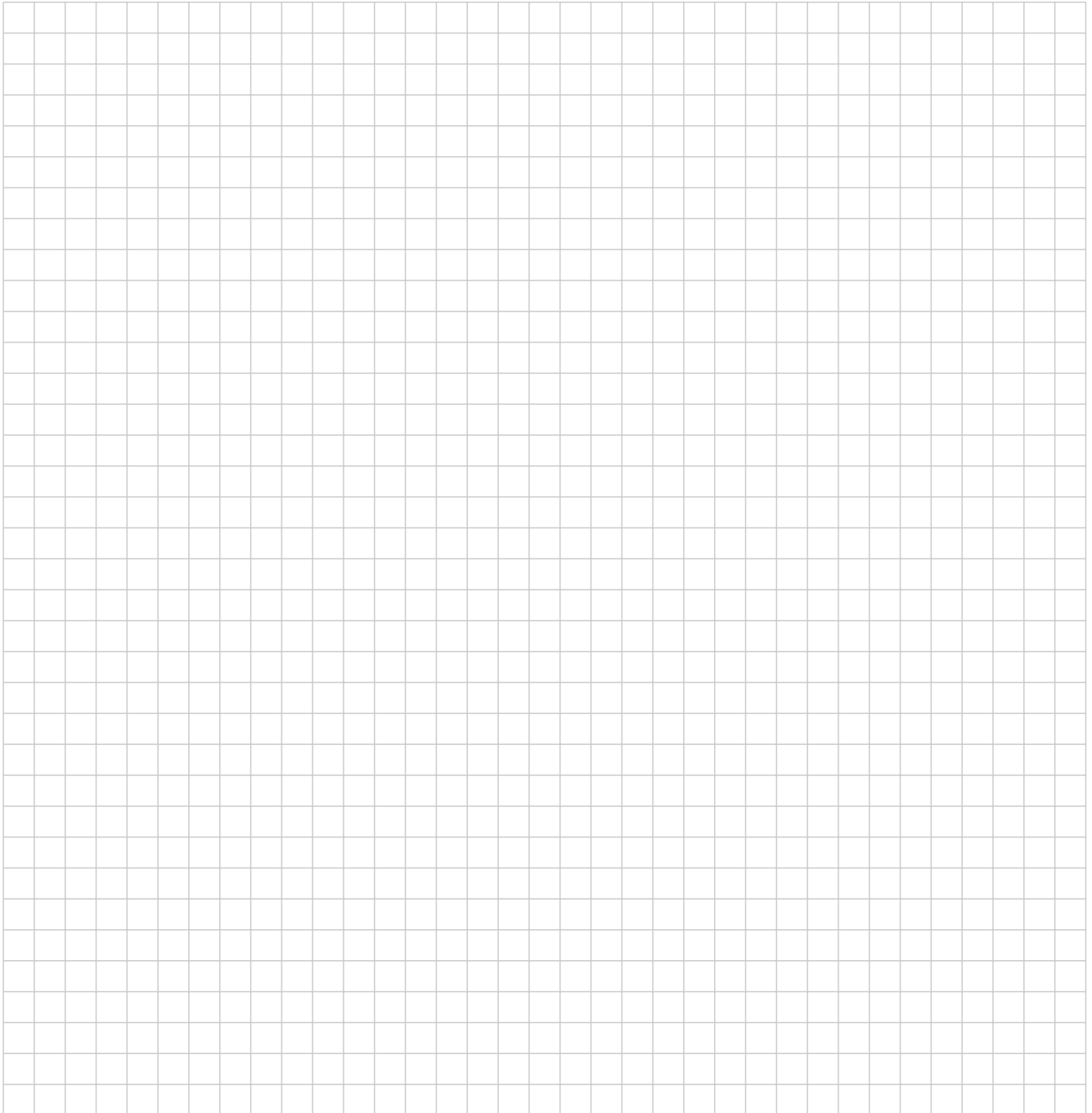
- 6p **6b** Were the participants who were treated rudely in the mass market willing to pay more for the product? Use hypothesis testing with a fixed significance level (of $\alpha = 0.05$) to determine the answer.



Dropped calls

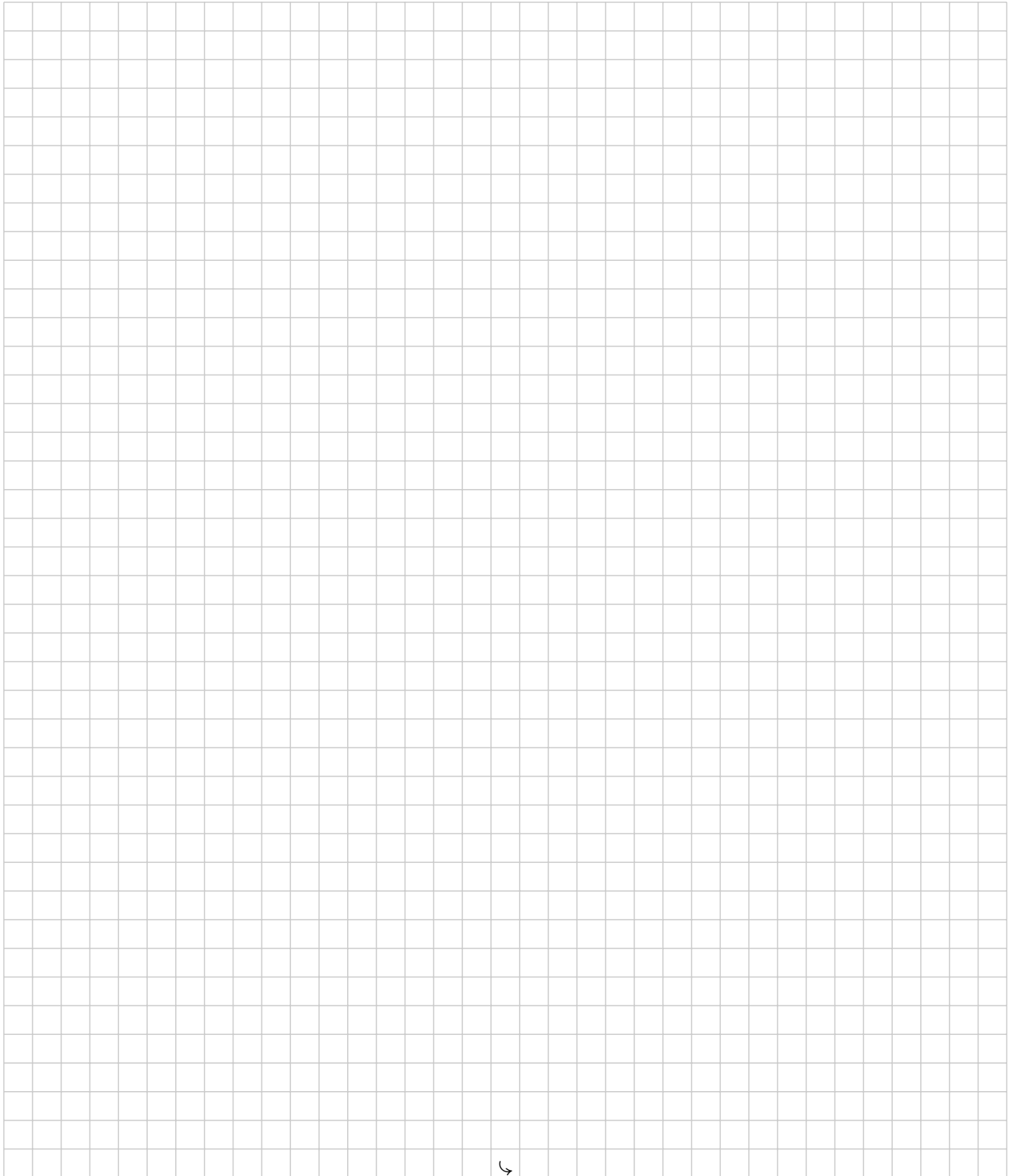
- 12p 7 Stephen wants to check whether we can use a Poisson distribution with an average of 2 calls per day to model the number of dropped calls on his cell phone. To do this, he records the number of dropped calls per day for the next 100 days. His observed counts of dropped calls are 11, 22, 28 and 39, for having 0, 1, 2, and ≥ 3 dropped calls, respectively.

Use a χ^2 -test goodness of fit test to test the hypothesis that the number of dropped calls follow a Poisson distribution.



Extra Space

- 8 Use the space below if you ran out of space for other questions. Clearly indicate which question you are answering.



A large grid of graph paper for extra space. The grid is 30 columns wide and 30 rows high. A small cursor icon is located at the bottom center of the grid, near the intersection of the 15th and 16th columns and the 30th row.



